TriADE report

2026-11-13 20:06 · TriADE 1.4.0

Methods

Acoustic-ejection MS runs (2 plate(s), 384-well format, unit-resolution data, both mode) were analysed with TriADE 1.4.0. The interval between ejections was fitted to the total ion current (2.500 s) and each ejection was assigned to its well (row-serpentine (A1→A24, B24→B1, ...)). For each well the product ion(s) [M+H]+, [M+Na]+, [M-H]- were searched within ±0.5 Da of the calculated m/z in every spectrum from 0.45 s before to 0.6 s after the ejection apex. An ion was considered validated when the M peak was ≥ 200,000 cps with a distinct M+1 peak ≥ 50% of the cutoff in at least 2 spectra, and the ejection signal was ≥ 2× the median of the same m/z in the wells sharing the same reagent. Wells were graded in five confidence levels: 5 Present (evidence ≥ 1,000,000 cps), 4 Likely present (≥ 200,000 cps), 3 Check spectrum (≥ 80,000 cps, or a reagent ion on the product ion, carryover ≥ 5× from the previous ejection, or only a secondary adduct), 2 Likely absent (raw signal > 100,000 cps) and 1 Not detected; wells that could not be measured were sent to HPLC-MS. The standard detection criteria of the program were used. HPLC-MS conversion ≥ 5% defined the presence of the product; wells without an HPLC-MS result were counted as not detected.

HPLC-MS triage

Worklist: 4: level 5 = present; levels 1-2 = absent — 308 wells to inject.

Scenario	HPLC injections	HPLC injections % of wells	HPLC time h	Time saved h	Net time saved h minus EchoMS	Mobile phase saved mL	False present	Products missed
0: everything to HPLC-MS (no triage)	768	100.0	64.0	0.0	0.0	0	0	0
1: Echo-MS only rules out level 1	649	84.5	54.1	9.9	9.3	357	0	8
2: Echo-MS only rules out levels 1-2	418	54.4	34.8	29.2	28.6	1050	0	34
3: level 5 = present; level 1 = absent	539	70.2	44.9	19.1	18.5	687	5	8
4: level 5 = present; levels 1-2 = absent	308	40.1	25.7	38.3	37.8	1380	5	34
5: levels 4-5 = present; levels 1-2 = absent	96	12.5	8.0	56.0	55.4	2016	40	34

Warnings: Plate 1: 17 ejection(s) flagged by the run QC (weak pulse or internal standard). | Plate 2: 15 ejection(s) flagged by the run QC (weak pulse or internal standard).

Plates

Plate	mzML	Wells	Scan range mz	Detected	Level 5 Present	Level 4 Likely present	Level 3 Check spectrum	Level 2 Likely absent	Level 1 Not detected	Level 0 Not measurable	Agreement %	Cohen kappa	Cohen kappa CI95
1	260616 Gustav 320-640 Da del 2500.mzML	384	320.0-640.0	137	37	92	48	121	86	0	86.5	0.717	0.646-0.785
2	260616 Gustav 290-640 Da del 2500.mzML	384	289.9-640.1	205	73	120	48	110	33	0	82.8	0.657	0.58-0.73

Plate 1 — Confidence levels

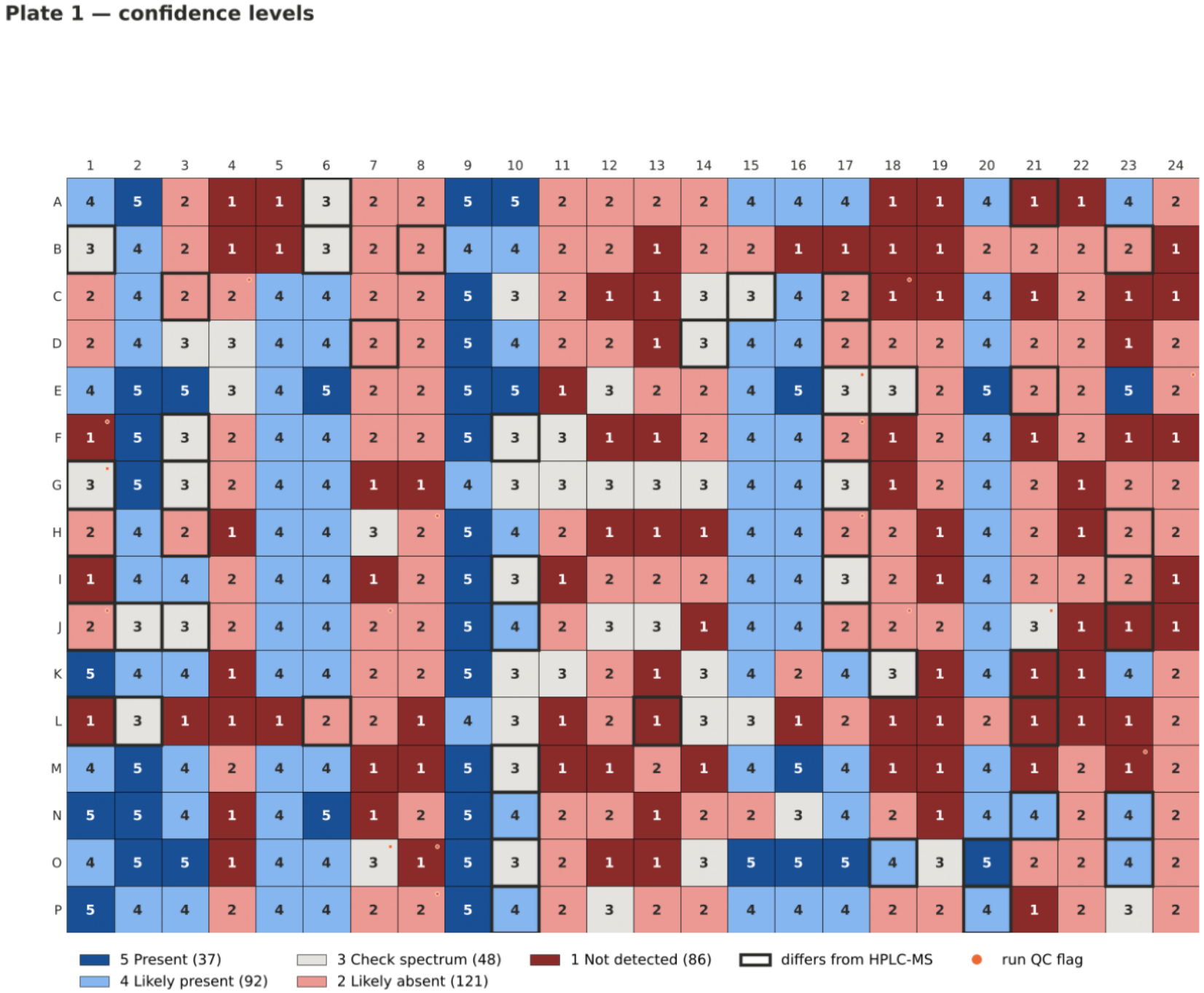


Plate 1 — Echo-MS levels and HPLC-MS conversion

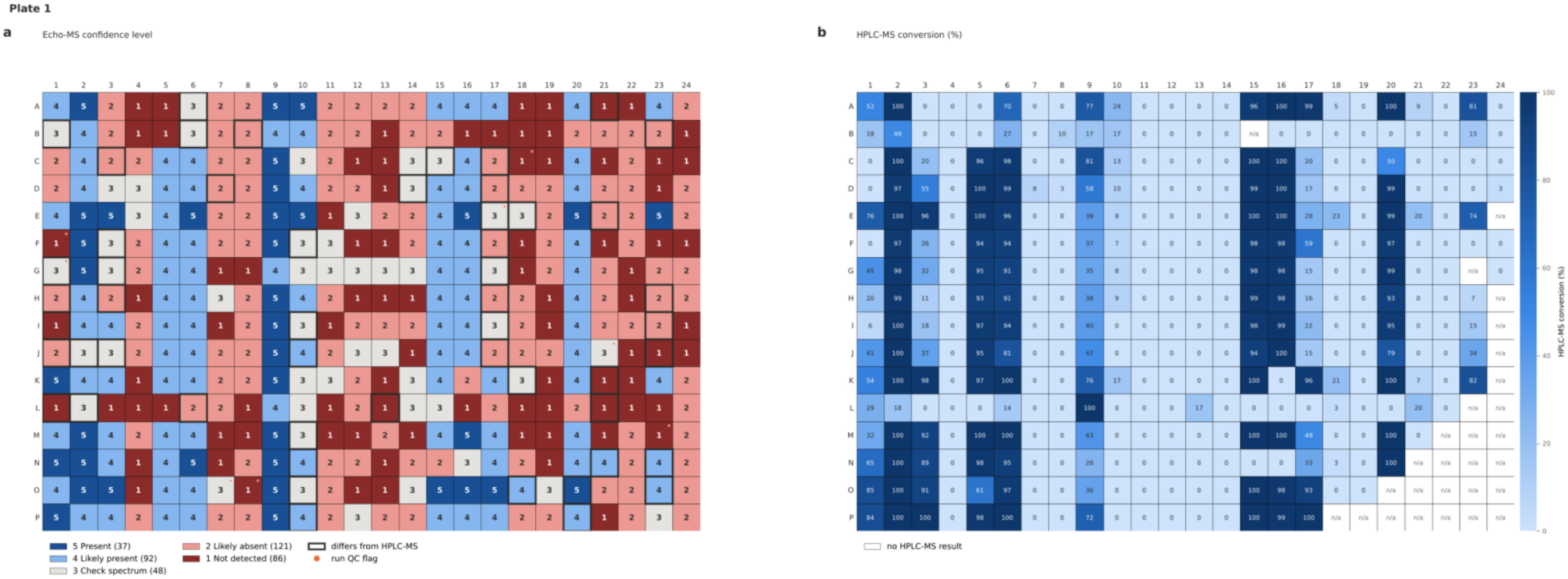
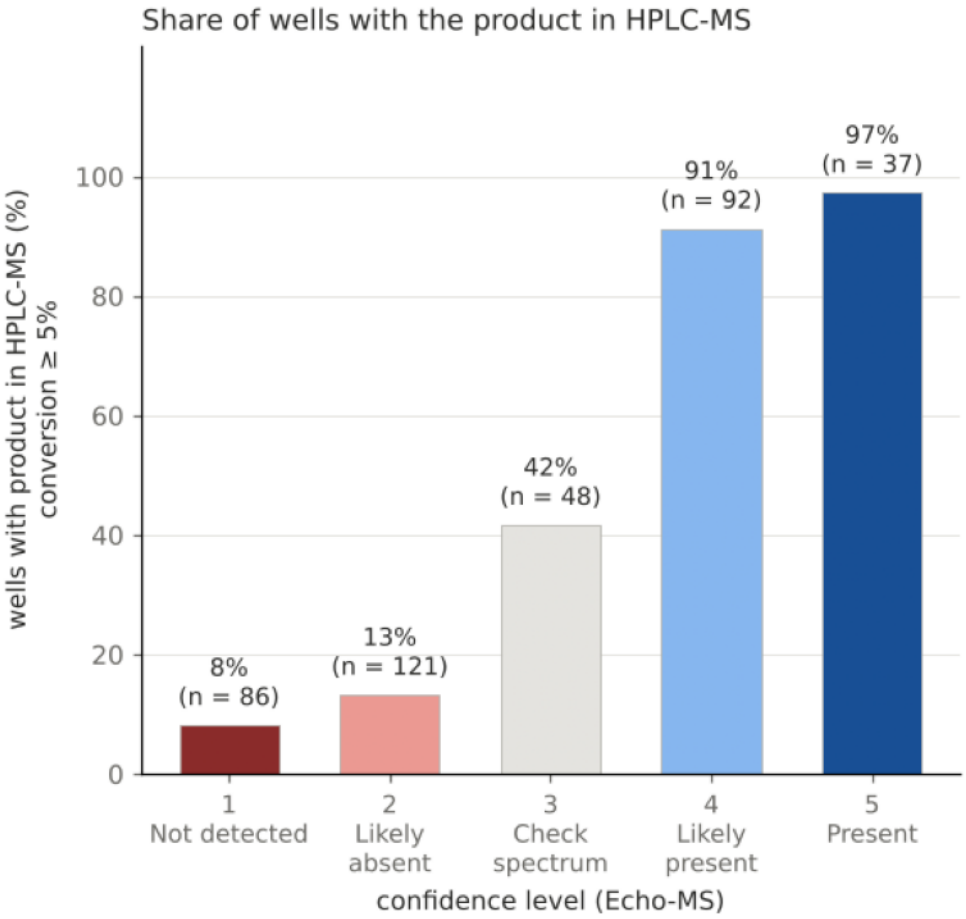


Plate 1 — confidence levels vs. HPLC-MS (384 wells): level 5 = 97% with product, level 1 = 8%

28 wells with no HPLC-MS signal are counted as not detected (no product).



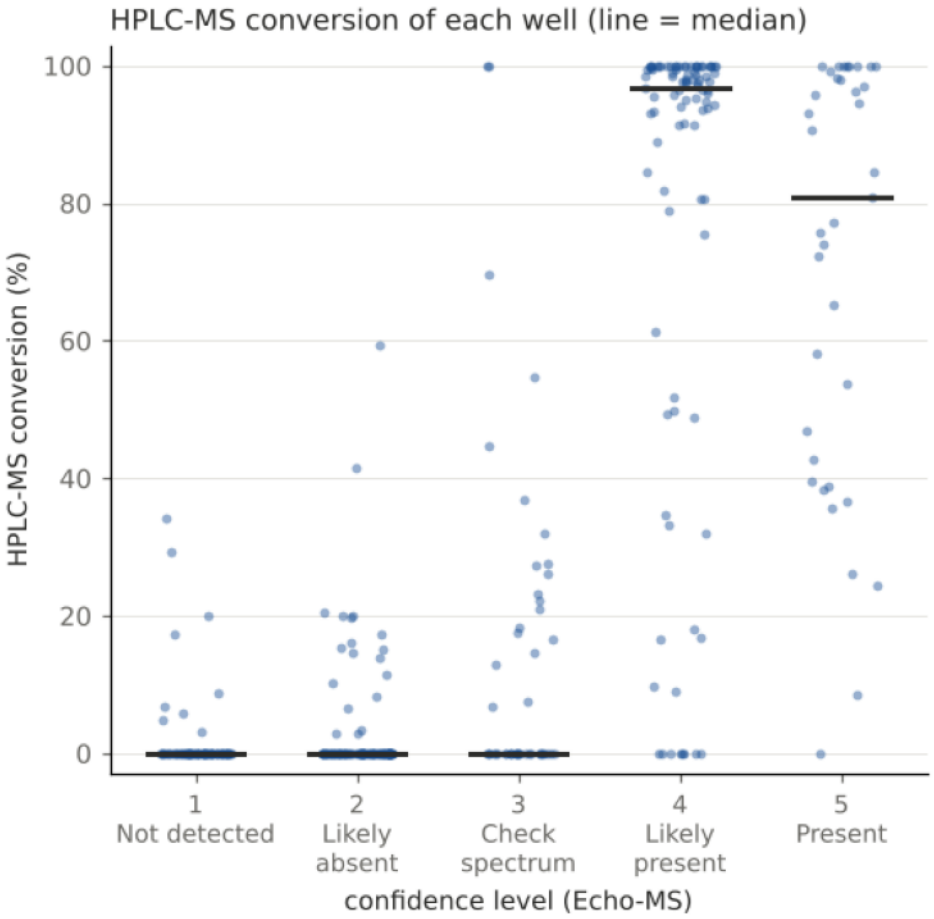


Plate 2 — Confidence levels

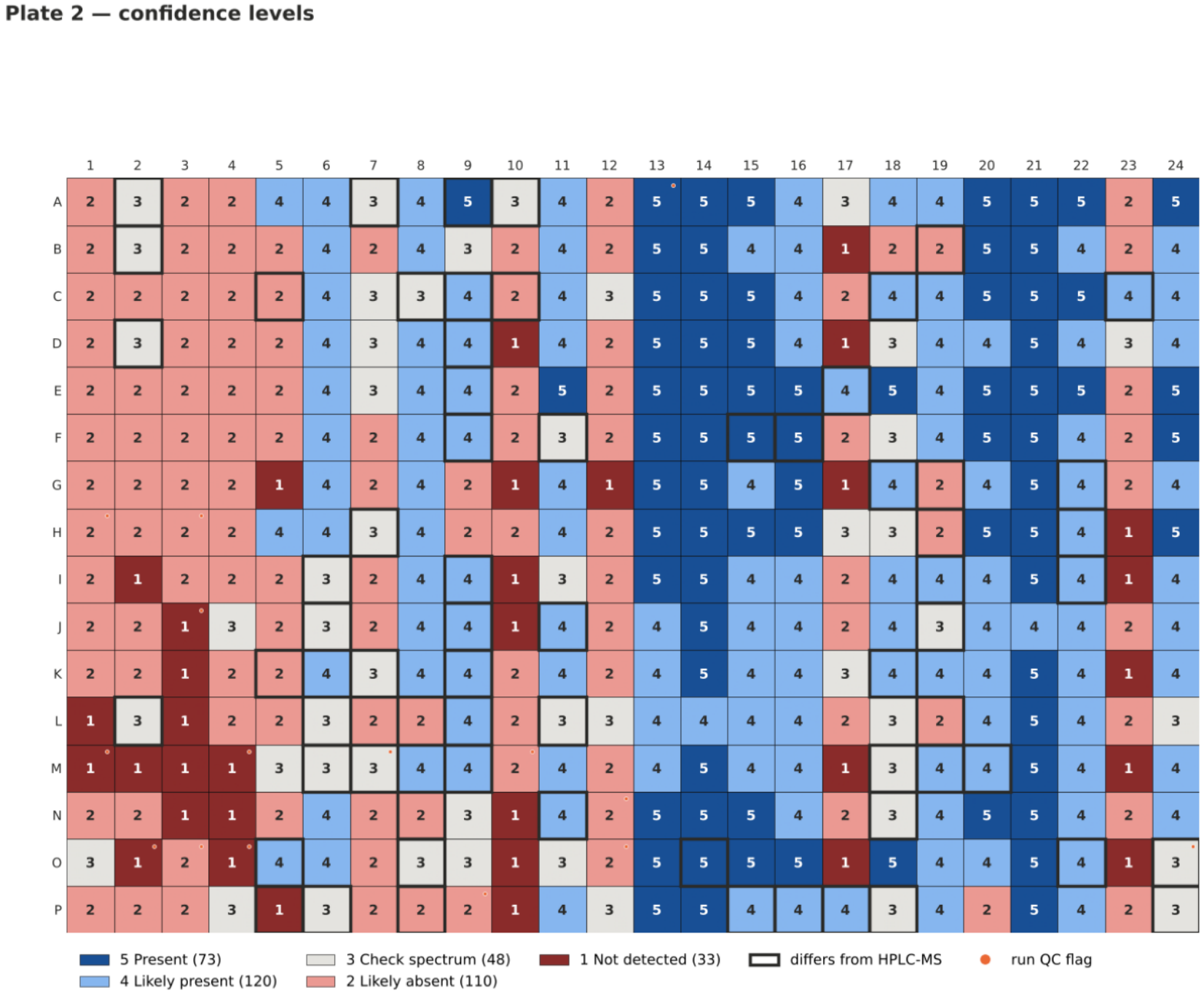


Plate 2 — Echo-MS levels and HPLC-MS conversion

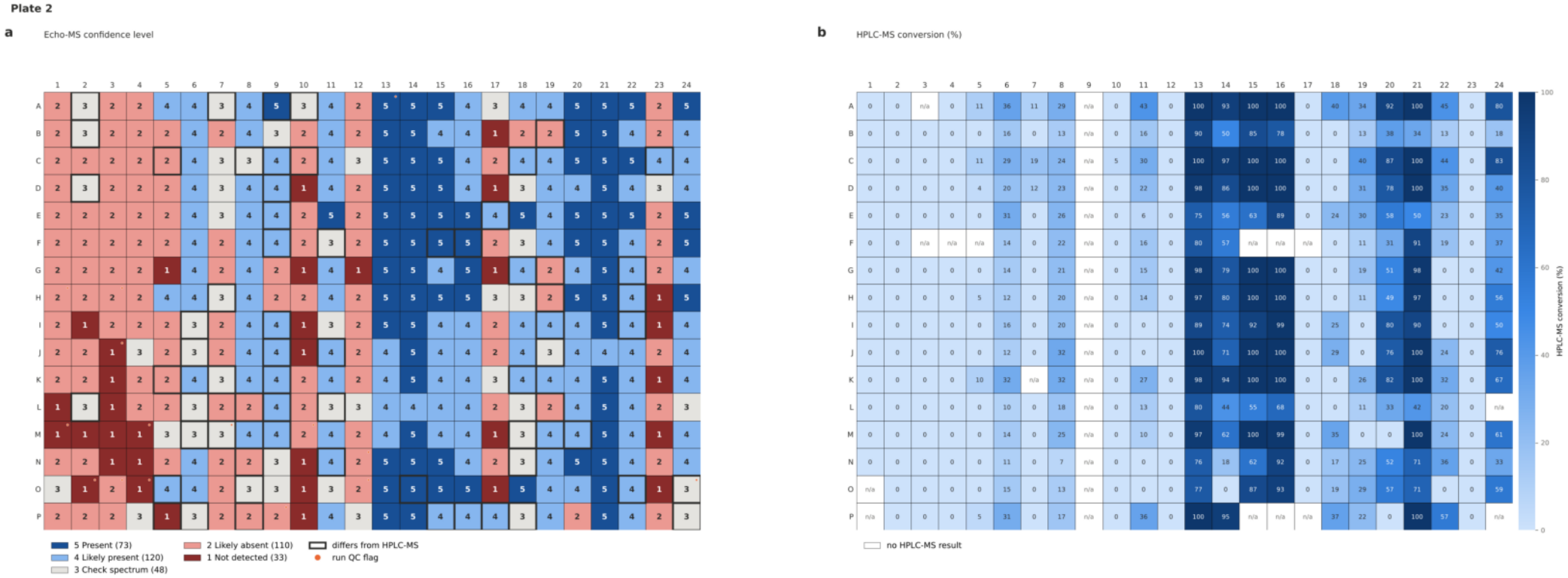


Plate 2 — Levels vs. HPLC-MS

Plate 2 — confidence levels vs. HPLC-MS (384 wells): level 5 = 95% with product, level 1 = 3%

15 wells with no HPLC-MS signal and 16 marked as indeterminate are counted as not detected (no product).

